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Option Return Predictability via Machine Learning: New Evidence From China

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ABSTRACT

We extend the literature on empirical asset pricing to the Chinese options market by building and analyzing a comprehensive set of return prediction factors using various machine learning methods. In contrast to previous studies for the US market, we emphasize the uniqueness of this emerging market, investigate daily hedging strategies to construct delta-neutral portfolios, and identify the most important characteristics for return prediction. Short-selling restrictions in China's financial market diminish the effectiveness of spot hedging, whereas delta-neutral portfolios based on futures hedging deliver substantial improvements in both annual returns and Sharpe ratios. Machine learning models not only outperform the IPCA benchmark, but also demonstrate strong generalization ability when applied to newly issued option contracts. The out-of-sample performance remains economically significant after accounting for transaction costs.

JEL Classification: G10, G12, G13, G14

1 | Introduction

Since the launch of the first option product in 2015, China's options market has been expanding rapidly. A prominent example is the SSE50 option, whose underlying asset is China's largest Exchange Traded Fund (ETF). According to the recent Stock Options Market Development Report published by the Shanghai Stock Exchange (SSE),[1] approximately 460 million SSE50 option contracts were traded in 2023 with a total transaction value of 12.13 trillion yuan (equivalent to around 1.71 trillion USD[2]), marking a more than tenfold increase compared to 2015. Currently, this emerging market continues to issue new products for investors to choose from. However, it is not simply the size and growth rate, but also the uniqueness of the Chinese options market that makes it particularly appealing for academic research. This specificity enables us to investigate questions that deepen our understanding of emerging markets and shed light on how financial systems operate in different institutional contexts. In particular, we identify the following unique features of the Chinese options market.

First, in terms of tradable products, China's exchange-traded financial options are primarily index options and ETF options with the temporary absence of individual stock options. By comparison, statistics show that single-stock options on the Chicago Board Options Exchange had an average daily trading volume of approximately 12.409 million contracts in 2023, comprising 76.6% of the total volume.[3] Existing research has confirmed the difference between index option and single-stock option in pricing dynamics and return realization (Broadie et al. 2009; Chambers et al. 2014). For instance, Santa-Clara and Saretto (2009) show that index options involve substantial bid-ask spreads and margin requirements, which are essential considerations for accurately estimating option returns. Constantinides et al. (2013) conclude that only four crisis-related factors effectively explain the cross-section of index option returns and that the leverage-adjusted returns on S&P Index options strongly contradict the predictions of Black and Scholes (1973) model. Against this background, we construct a wide range of pricing factors to investigate return predictability in the Chinese options market and leverage machine learning methods to model complicated relationships between numerous factors and expected returns.

Second, in terms of operational environment, China's financial regulator enforces strict restrictions on short sales in the spot markets. These restrictions were not relieved until March 2010, when the China Securities Regulatory Commission (CSRC) permitted a limited number of brokerage firms to short-sell a constrained range of stocks from a designated list (Gao and Ding 2019). Although there is no universal agreement, some studies suggest that short-selling contributes to price discovery and improves market efficiency (Saffi and Sigurdsson 2011). Growing empirical evidence also indicates that short-selling costs significantly affect option prices by widened bid-ask spread (Evans et al. 2009; Ramachandran and Tayal 2021), distorted implied volatility (Lin and Lu 2016) and apparent put-call parity violation (Atmaz and Basak 2019). In comparison, China's derivatives market has a more developed short-selling mechanism. Futures prices more accurately reflect the true supply-demand relationships and can replace misleading spot prices for more precise asset pricing. In this context, we derive implied volatility surface based on spot prices and future prices, respectively. Delta-neutral portfolios are subsequently constructed using either spot hedging or futures hedging strategies to compare their differences in realized returns and factor importance.

Third, in terms of development level, the Chinese options market is still in the development phase rather than the mature phase. From the perspective of investor structure, China's options market, different from its stock market, is dominated by institutional investors. By the end of 2023, institutional investors contribute the 68.77% of the total option trading volume in SSE, while individual investors account for the other 31.23%.[4] By contrast, the U.S. options market has recently experienced a surge in retail trading. de Silva et al. (2023) provide evidence that individual investors in US tend to purchase options in a highly concentrated manner in anticipation of volatility spikes. Differences in behavior patterns, information processing, and trading strategies among different types of investors can influence the importance of financial factors (Vayanos and Woolley 2013; Liu et al. 2019). Therefore, we evaluate factor effectiveness by analyzing the interpretability of machine learning forecasts and examine what kinds of characteristics, in such a market, are significant to option pricing. From the perspective of product issuance, China's financial market continues to introduce new options products. For instance, in May 2023, CSRC initiated the listing of STAR50 ETF options[5] to diversify risk management tools and support capital market development. How to effectively price newly issued derivatives has garnered extensive attention from both industry and academia. Against this background, we explore whether machine learning models trained with historical data can exhibit strong predictive power for out-of-sample option contracts.

As a crucial tool for explaining asset prices, financial factors hold a pivotal position in portfolio construction and risk management (Li et al. 2023). Option factors indicating excess returns have received increasing attention (Liu et al. 2024). A plethora of variables such as demand pressure (Bollen and Whaley 2004; Garleanu et al. 2008), stock volatility (Cao and Han 2013), gambling characteristics of the underlying stock (Byun and Kim 2016), volatility and jump risk (Guo and Lin 2020), stock liquidity (Kanne et al. 2023) and dynamic conditional skewness (Liang and Du 2024) are found to be valuable predictors of option-based asset returns. At present, a comprehensive factor database for Chinese options market remains unavailable. To fill this gap, we contribute to empirical asset pricing research by constructing a unique factor library, which include 119 option characteristics and 57 technical indicators in total. First, we build a set of factors using the same methodology applied to US market. Second, we adapt some of these factors to better align with the features of Chinese option market.[6] Finally, we include technical factors to capture the dynamics of underlying assets, such as price volatility and momentum.

Given the rapid and unique evolution of the Chinese options market, we posit that highly adaptable methods are essential to capture its complexities. Thus, we employ various machine learning techniques, whose application in finance, particularly in option pricing and return prediction, has recently gained substantial momentum and delivered promising insights. We build on recent advancements by Bali et al. (2023), who applied a suite of machine learning methods to examine return predictability in the US options market. Their findings show that machine learning improves accuracy in option pricing and enhances portfolio performance, particularly through sophisticated models that capture complex nonlinear interactions among predictors. Whether these results can be extended to the Chinese options market remains uncertain. However, given its uniqueness, such an emerging market represents a promising area to further explore machine learning's ability in modeling intricate pricing dynamics and return predictability.

By comparing the predictive power of different methods, we find that machine learning models outperform IPCA benchmark (Büchner and Kelly 2022) by generating more profitable portfolios and that nonlinear models span larger efficient frontiers than linear models. The advantages of nonlinear models stem from their ability to capture complex dynamics among variables, such as abrupt changes and non-monotonic relationships between predictors and expected returns. Tree-based models, due to their flexibility in modeling nonlinear relationships, stand out as an effective tool for predicting delta-neutral returns. Specifically, random forest achieves an annualized return of 69.6% for long-short portfolios under futures hedging strategy. The predictability of option returns is largely driven by informational frictions and mispricing in the options market. The robustness of nonlinear machine learning's advantages can be validated across various testing scenarios. They can still obtain optimal risk-adjusted returns even after accounting for transaction costs, incorporating additional factors, and extending to newly issued products.

We further assess factor importance using the efficient model explanation framework SHAP (Lundberg and Lee 2017). The result shows that option characteristics related to contract information and risk have strong explanatory power for price variation, while technical indicators, reflecting price movements of the underlying asset, are less important in explaining delta-neutral returns. Comparative analysis reveals that factor importance depends on the specific hedging strategy. Spot hedging places greater emphasis on deviations caused by short-selling constraints, whereas futures hedging attaches more importance to asset illiquidity. Compared with the findings of Bali et al. (2023) in the US market, we find that factors related to put-call parity violations and jump risk are more significant in the Chinese options market, while factors associated with retail trading exert less influence. These differences highlight the unique features of the Chinese options market, particularly in terms of short-selling constraints, asset liquidity, and the composition of market participants.

Since the directional impact of stock prices can be eliminated through frequent hedging procedures (Tian and Wu 2023), this paper focuses on risks that are inherently nonlinear and likely to interact with each other in complex ways. Following the methodology of Gu et al. (2020), we investigate how nonlinearities and interactions among a large set of option characteristics influence model predictions. Our marginal association analysis reveals nonlinear relationships between expected returns and key option characteristics, illustrating how model predictions vary with changes in factors like implied volatility and illiquidity measures. Furthermore, our examination of interaction effects highlights the different impact of demand pressure on option prices under different levels of conditional variables. Together, these findings underscore the importance of accounting for extensive nonlinearities and interactions among numerous predictors when predicting option-based asset returns.

Our work contributes to the booming literature that applies machine learning and model explanation techniques to asset pricing, particularly in the context of large datasets containing numerous characteristics (Leippold et al. 2022). Despite the successful application of machine learning in the US options market (Bali et al. 2023), the differences in factor effectiveness between China and US stock markets (Jansen et al. 2021; Leippold et al. 2022) suggest that conclusions drawn from developed financial markets may not directly apply to China and require careful reassessment in light of its uniqueness. To address this gap, we examine the distinctive features of Chinese financial market through various machine learning techniques. We not only evaluate factor effectiveness in the Chinese options market, but also integrate multi-factor quantamental investing strategies based on model predictions, offering novel insights into option return dynamics.

This paper also contributes to the literature on option pricing factors and anomalies. Although many sophisticated models have been proposed for option pricing (Heston 1993; Merton 1976), they still struggle to make sufficiently accurate predictions. Recent studies suggest that option pricing should consider more diversified factors beyond exposure to underlying assets (Garleanu et al. 2008; Büchner and Kelly 2022). These successful applications mainly focus on the US options market, leaving a gap in understanding the emerging derivatives market from a multi-factor perspective. In contrast to the existing literature, our main contributions are three-fold. First, we leverage machine learning to predict next-day delta-neutral returns by capturing complex relationships in large-scale, high-frequency data, and identify differences in factor effectiveness between the Chinese and US options markets, highlighting how variations in trading restrictions and investor composition shape the factor structures underlying option return predictability. Second, we identify the key pricing factors that drive return predictability in the Chinese options market and reveal the critical role of nonlinear and interactive relationships among these characteristics in predicting option returns. Third, we conduct a detailed comparison between spot-based and futures-based delta hedging strategies and demonstrate that hedging choices significantly affect option return predictability and the relative significance of predictive factors, providing new insights into how trading strategies shape option return predictability.

The remainder of the paper is organized as follows. Section 2 describes the variables and methodologies used for prediction. Section 3 explores whether predictability translates into portfolio gains, compares different hedging strategies and evaluates out-of-sample performance. Section 4 identifies the key factors and provides evidence on their influence on predictions. Section 5 concludes this paper. Detailed methods and additional results are in Supporting Information S1: Appendix.

2 | Methodology

2.1 | Forecasting Framework

Option pricing can be conceptualized as a supervised learning problem, wherein the objective is to estimate parameters for the following functional form:

$$r_{i,s,t+1} = f(x_{i,s,t};\theta) + \epsilon_{i,s,t+1}, \tag{1}$$

where $f(\cdot;\theta)$ denotes a machine learning function parameterized by θ. Here, $r_{i,s,t+1}$ represents the return of the i-th option contract at time $t+1$ for the underlying asset s, $x_{i,s,t}$ encapsulates the asset characteristics at time t, and $\epsilon_{i,s,t+1}$ is the error term.

To specify the functional form of f, the factor data is partitioned into mutually exclusive training and testing sets. Model parameters are calibrated using the training set, and their predictive capacity is evaluated on the testing set. We employ a cross-validation approach for parameter tuning, optimizing data utilization while ensuring the robustness of selected models through repeated evaluations. Similar to Bali et al. (2023), models are periodically retrained at specified intervals with most recent factor-return data.[7]

Hyperparameters are determined by random search method, as it is more effective than grid search in high-dimensional parameter spaces (Bergstra and Bengio 2012; James et al. 2013). We construct portfolios based on their predictions and use metrics such as annual return, Sharpe ratio and information coefficients[8] to evaluate model performance. We construct a long portfolio using assets with the top 10% predicted returns and a short portfolio using those with the bottom 10%. Due to the high dimensionality of the characteristics, we employ the SHAP method (Lundberg and Lee 2017), a general framework and computationally efficient tool for model explanation, to quantify predictors' contribution.

Drawing on existing literature that combines machine learning with asset pricing (Gu et al. 2020), this study selects four linear machine learning models and five nonlinear machine learning models. The linear models include Lasso regression, Ridge regression, Elastic Net (ENet), and Partial Least Squares regression (PLS). The nonlinear models include Support Vector Machines (SVM), Deep Feedforward Networks (DFN), Random Forest (RF), Gradient Boosting Decision Tree (GBDT), and Extreme Gradient Boosting Tree (XGBoost). Following Rapach et al. (2010), this paper separately aggregates linear and nonlinear models through ensemble learning to investigate whether different models could complement each other in prediction. A simple and intuitive approach is adopted, assigning equal weights to the aggregated base models.

2.2 | Data Description

This paper utilizes all exchange traded financial options listed in Chinese derivatives market from February 2015 to July 2024 as the research sample. Financial options are selected because of their larger trading volume and open interest compared to commodity options.[9] Besides, China's on-exchange financial options are exclusively European-style, which eliminates the need to consider early exercise, thereby simplifying pricing through data-driven models. Option-based portfolios require more frequent adjustments based on market conditions to mitigate the risk of holding unfavorable positions. Accordingly, we use daily frequency data to evaluate the performance of machine learning models in high-frequency trading, which distinguishes this study from previous research.

Consistent with Boulatov et al. (2022), we retain option contracts with relatively low nominal prices. Since option returns inherently contain significant noise, particularly for those with low trading activity, we adopt several filtering methods to ensure prices truly reflect market information (Büchner and Kelly 2022; Bali et al. 2023): (1) Options with missing or abnormal implied volatility values are excluded; (2) Options with zero trading volume over past seven calendar days are disregarded, and only options with non-zero open interest are considered; (3) Options with a bid price of zero or whose ask price is less than or equal to the bid price are removed to minimize microstructural noise; (4) Options expiring within 2 weeks are ruled out; (5) The non-arbitrage condition of option prices is verified for each underlying asset, requiring that, given the same time to maturity and underlying asset, call (put) options with lower strike prices have higher (lower) prices.

2.3 | Option Return

The primary objective of this paper is to generate excess returns by constructing a delta-neutral portfolio on a daily basis. Following Bakshi and Kapadia (2003), the delta-neutral return can be conceptualized as the value change of a portfolio consisting of a long position in the option and a hedging position in underlying asset. Such portfolio design is locally immune to price fluctuations of the underlying asset (Tian and Wu 2023).

To accurately characterize returns, let the time interval from t to $t+\tau$ be divided into discrete time points $T = \{t = t_0 < \cdots < t_N = t + \tau\}$, and assume that portfolio remains delta-neutral at each of the N discrete time points $t_n, n = 0, 1, \cdots, N-1$. The delta-hedged option gain over interval $[t, t+\tau]$ can be expressed by:

$$\begin{aligned}\Pi(t, t+\tau) = V_{t+\tau} - V_t - \sum_{n=0}^{N-1} \Delta_{V,t_n} \times [S(t_{n+1}) - S(t_n)] \\ - \sum_{n=0}^{N-1} \frac{a_n r_n}{365} [V(t_n) - \Delta_{V,t_n} S(t_n)],\end{aligned} \tag{2}$$

where V_t represents the option price at time t, r_n denotes the risk-free rate at time t_n, and a_n represents the number of calendar days between two consecutive hedging dates t_n and t_{n+1}. Δ_{V,t_n} denotes the delta value of the option. S refers to the price of underlying asset, which is spot asset or futures depending on the specific hedging strategy. Finally, following the methodology of Zhan et al. (2022), the option return is defined as follows:

$$r_{t,t+\tau} = \frac{\Pi(t, t+\tau)}{|\Delta_t S_t - V_t|}. \tag{3}$$

2.4 | Variable Construction

We construct three levels of factors to describe option characteristics. The first comprises asset-level factors that distinguish options with different underlying assets, such as the put-call ratio (*pcratio*). The second includes bucket-level factors that differentiate options belonging to different buckets,[10] such as *bvol*. The third involves contract-level factors that characterize individual options, such as the Greeks and implied volatility. Based on this taxonomy, we build up a comprehensive option factor data set consisting of 119 signals, whose construction is described in more detail in Supporting Information S1: Appendix D. Following Jensen et al. (2023), option factors are further categorized into six primary groups based on their implied information: Contract, Risk, Illiquidity, Frictions, Past Prices, and Informed Trading. To investigate whether characteristics about underlying asset is helpful to predicting delta-neutral returns, we also construct 57 technical indicators to capture price dynamics of the underlying asset. Details about the construction and classification of technical indicators are provided in Supporting Information S1: Appendix E.

To compare different hedging strategies, we consider both spot prices and futures prices to derive two types of implied volatility surfaces. Although realized returns are calculated with the prices of options and underlying assets, information embedded in implied volatility surfaces is very crucial for constructing option characteristics and, accordingly, influences both model prediction and portfolio composition. We finally build up a comprehensive "factor-return" data set by aligning the characteristics from period $t-1$ with the corresponding actual returns at time t. To mitigate potential prediction biases, we adjust factors with substantial differences in magnitude to eliminate scale-related distortions during their construction and normalize them before model input.

2.5 | Summary Statistics

After applying previously introduced filtering methods, the final sample comprises 16,594 option contracts spanning 12 distinct underlying assets with a total number of approximately 993,000 observations. Table 1 presents the descriptive statistics calculated with future prices, highlighting key characteristics across different option types. The average return of delta-neutral options is −0.096% and the mean implied volatility is 21.3%. The median time to maturity is 79 calendar days. The average moneyness, which measures the relative size between strike price and the current price of underlying asset, is 1.008. Within this data set, 50.2% are call options and the other 49.8% are put. Both call and put options exhibit negative mean and median returns. Delta-neutral returns display pronounced skewness and kurtosis, and the Jarque-Bera test strongly rejects the null hypothesis of normality. Summary statistics based on spot prices are presented in Supporting Information S1: Appendix C. A comparison of statistics derived from different hedging strategies reveals substantial discrepancies in implied volatility. Since the construction of option characteristics largely depends on information extracted from the implied volatility surface, the resulting datasets display divergent factor values, which in turn can lead to variations in factor importance, portfolio composition, and return realization.

3 | Empirical Results

3.1 | Model Evaluation

We start by exploring the empirical performance of machine learning-driven multi-factor investment in the Chinese options market. Since the excess return of delta-neutral portfolios is theoretically zero under the Black and Scholes (1973) framework, it is crucial to compare machine learning algorithms with other benchmark models. Due to the finite lifespan and drifting moneyness of options, it is infeasible to directly construct a Capital Asset Pricing Model by estimating beta exposures to market indices. To address these challenges, Büchner and Kelly (2022) proposes the Instrumented Principal Components Analysis (IPCA). IPCA captures time-varying loadings on latent factors associated with option prices and has effectively explained monthly return variation in S&P 500 index options. In this study, we employ IPCA as a benchmark for comparison with machine learning algorithms. However, due to its sensitivity to multicollinearity, only the factors used in the original paper is included as inputs to the IPCA model.[11]

Table 2 presents the performance evaluation of different models. Futures hedging strategy is adopted and training data spans

TABLE 1 | Summary statistics based on futures prices.

	Mean	Sd	10-Pct	Median	90-Pct	Skew	Kurt
Panel A: All options							
Return	−0.10%	1.49%	−0.69%	−0.04%	0.62%	−7.778	278.601
Maturity	106.304	79.931	24	79	222	0.987	0.251
Moneyness	1.008	0.126	0.860	0.998	1.170	0.406	0.507
Implied volatility	0.213	0.084	0.147	0.203	0.298	2.240	18.956
Absolute delta	0.490	0.329	0.047	0.485	0.941	0.041	−1.405
Panel B: Call options							
Return	−0.10%	1.58%	−0.75%	−0.05%	0.68%	−10.227	213.336
Maturity	106.581	80.046	25	79	222	0.981	0.236
Moneyness	1.011	0.125	0.864	1.001	1.174	0.386	0.434
Implied volatility	0.213	0.082	0.147	0.203	0.297	2.046	14.581
Absolute delta	0.503	0.329	0.048	0.514	0.942	−0.032	−1.408
Panel C: Put options							
Return	−0.09%	1.40%	−0.63%	−0.04%	0.58%	−4.098	375.061
Maturity	106.024	79.814	24	79	222	0.993	0.266
Moneyness	1.004	0.126	0.857	0.994	1.167	0.427	0.586
Implied volatility	0.213	0.086	0.146	0.202	0.299	2.407	23.513
Absolute delta	0.477	0.328	0.046	0.456	0.940	0.116	−1.387

Note: This table reports descriptive statistics for delta-hedged option returns based on futures prices. The short-selling mechanism in the Chinese futures market is relatively well-developed, whereas the spot market imposes stricter restrictions on short selling. Consequently, futures prices can more accurately reflect market expectations and supply-demand dynamics, and can be utilized for more precise estimation of implied volatility surfaces. The columns, from left to right, report the mean, standard deviation (Sd.), 10th percentile (10-Pct), median, 90th percentile (90-Pct), skewness (Skew), and kurtosis (Kurt). Delta-hedged option returns are measured over a one-unit time period, with delta-hedging performed daily using spot assets. Panels B and C present the descriptive statistics for call and put options, respectively.

across 80 trading days. Panels A and B of Table 2 report the annual return and Sharpe ratio of different portfolios, while panel C and D present the mean, standard deviation, and t-statistics of Newey and West (1987) test for two types of information coefficients.

In terms of investment performance, both linear and nonlinear machine learning models outperform IPCA benchmark, demonstrating the superior effectiveness of machine learning in multi-factor quantamental investing. Furthermore, the information coefficients of machine learning models are significantly greater than zero, while the IPCA used for comparison has noticeably smaller average values and t-statistics to significantly reject null hypothesis. In addition, nonlinear models outperform their linear counterparts by learning more complex patterns that help construct more profitable portfolios. Among linear machine learning models, Lasso regression delivers the highest annual returns, while linear ensemble methods can achieve higher Sharpe ratios. For nonlinear models, the best performance is primarily driven by RF, GBDT and XGBoost, which is consistent with prior research suggesting that tree-based models are particularly well-suited for tabular data (Arik and Pfister 2021). Overall, the most favorable investment outcomes are consistently observed in nonlinear models, highlighting their stronger representational capacity in high-dimensional settings and their potential to yield superior out-of-sample performance (Gu et al. 2020; Bali et al. 2023).

Following Cong et al. (2025), we visualize the mean-variance efficient frontiers spanned by constructed decile portfolios. Figure 1 displays the efficient frontiers of various machine learning models and the IPCA benchmark, each represented with distinct shapes and colors. For clarity, segments of the curves below the minimum variance point are omitted. Frontiers generated by nonlinear models are marked with asterisks, while the IPCA benchmark is shown in gray and ensemble models in red. Notably, the frontiers constructed by machine learning models dominate that of the benchmark, indicating greater portfolio efficiency. In particular, nonlinear models such as RF and GBDT produce frontiers that are further shifted toward the top-left corner of the mean-variance diagram, suggesting improved risk-adjusted performance. These visual findings reinforce the results reported in Table 2, and further demonstrate that nonlinear models exhibit stronger capabilities in constructing more profitable portfolios and expanding the efficient frontier.

3.2 | Different Hedging Strategies

When constructing delta-neutral portfolios, investors need to hedge against delta risk by buying or selling the underlying asset. However, investors are unable to conveniently short sell in spot market due to relevant restrictions, which could lead to overestimated spot prices, incomplete hedging and unexpected risk exposures. As a result, alternative hedging strategies need to be considered. In this subsection, we construct delta-neutral portfolios using either spot hedging or future hedging strategy[12], and compare their investment performance to investigate the impact of short sales restrictions in the Chinese options market. When the price of underlying assets truly reflect market expectation and demand-supply balance, the implied volatility derived from paired put and call options should be consistent (Taleb 1997).

TABLE 2 | Machine learning model evaluation by investment performance and information coefficients.

	Lasso	Ridge	ENet	PLS	L-En	SVR	RF	GBDT	XGBoost	DFN	N-En	IPCA
Panel A: Annual return												
Long	0.423	0.101	0.378	0.196	0.403	0.366	**0.469**	0.378	0.418	0.020	0.452	0.067
Short	0.705	0.392	0.624	0.604	0.640	0.505	0.947	**0.960**	0.830	0.739	0.854	0.076
Long-Short	0.562	0.242	0.500	0.389	0.520	0.437	**0.695**	0.648	0.614	0.337	0.645	0.073
Panel B: Sharpe ratio												
Long	3.810	0.852	3.701	2.238	3.957	4.107	**5.403**	3.773	4.333	0.229	4.771	1.233
Short	5.940	3.307	5.289	4.418	5.741	4.837	6.261	6.023	**6.751**	5.972	6.449	1.264
Long-Short	7.770	2.957	7.485	5.712	8.004	7.585	8.811	7.953	7.942	5.156	**9.089**	2.971
Panel C: Information coefficients												
Mean	0.095	0.044	0.087	0.097	0.099	0.111	**0.148**	0.135	0.127	0.077	0.140	0.003
Std.	0.208	0.229	0.209	0.263	0.213	0.232	0.253	**0.266**	0.227	0.242	0.242	0.116
t-statistics	20.22	8.44	18.38	15.92	20.50	20.96	**25.35**	22.18	24.46	14.28	25.18	1.10
Panel D: Rank information coefficients												
Mean	0.098	0.047	0.089	0.116	0.105	0.131	**0.177**	0.152	0.139	0.088	0.160	0.008
Std.	0.221	**0.243**	0.221	0.239	0.218	0.223	0.230	0.227	0.222	0.231	0.222	0.194
t-statistics	19.24	8.49	17.57	21.22	20.98	26.02	**34.17**	29.54	27.61	17.05	31.78	1.63

Note: This table reports the performance evaluation about different machine learning models. Delta-neutral portfolios are constructed in terms of predicted returns and hedged by futures. L-En and N-En represent linear and nonlinear ensemble models, respectively. Panel A and Panel B report the annualized return and Sharpe ratio of different portfolios. Panel C and Panel D report mean, standard deviation (Std.) and Newey-West t-statistics of information coefficients. Maximum value in each row is highlighted in bold.

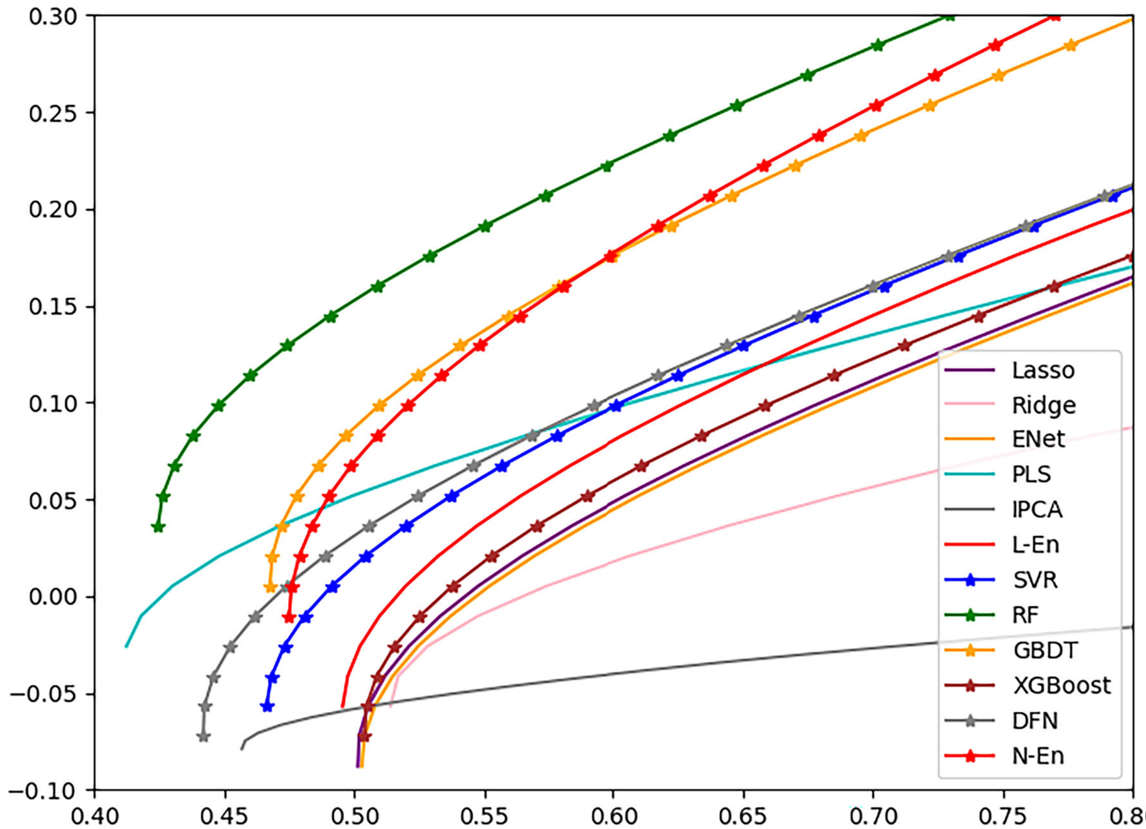

FIGURE 1 | Efficient frontiers based on different models. *Note:* This figure presents the mean-variance efficient frontiers constructed from different models. To differentiate model types, efficient frontiers corresponding to nonlinear models are additionally annotated with asterisks, while different colors are used to represent each model. The IPCA benchmark and the ensemble models are highlighted by gray and red solid lines, respectively. For clarity, segments of the curves lying below the minimum variance point are omitted. Notably, the efficient frontiers spanned by nonlinear machine learning models shift towards the top-left corner, indicating their ability to generate more profitable portfolios with lower risk.

The distinction between spot and futures hedging extends beyond differences on implied volatility surfaces, which also influences other option characteristics[13] and affects portfolios construction.

Table 3 reports the investment performance of long-short portfolios under two hedging strategies. The *diff* columns represent the difference in performance, calculated by subtracting the results of spot-hedged portfolios from those of futures-hedged portfolios. Performance metrics presented from left to right are annual return, Sharpe ratio, and maximum drawdown. As shown in Table 3, the futures hedging strategy significantly enhances both the annualized return and Sharpe ratio for most models. The return improvement is statistically significant at the 1% level for the majority of models, with a few exceptions including IPCA, Ridge regression, and DFN. Regarding maximum drawdown, spot-hedged portfolios outperform in only three cases (PLS, Ridge regression, and IPCA), suggesting that higher returns from futures hedging do not necessarily come at the cost of greater risk. Overall, these results highlight the superiority of the futures hedging strategy in terms of both return enhancement and risk control. It underscores the importance for investors to tailor investing strategies to the specific constraints and characteristics of the trading environment to optimize portfolio performance.

3.3 | Marginal Effect of Technical Indicators and High-Frequency Signals

In this subsection, we incorporate additional factors to investigate their impact on model predictions. Specifically, we introduce 57 technical indicators to assess whether characteristics of the underlying assets significantly influence option pricing. Given the well-documented importance of rolling window length for factor effectiveness and stability, we conduct independent single-factor tests during the construction process to ensure optimal selection and indicator quality. The results indicate that these technical indicators possess strong predictive power with respect to the price dynamics of the underlying assets. For example, a long-short portfolio constructed based on the *returnreverse* indicator, which captures price behavior following extreme movements and adjusts for historical volatility, generates a statistically significant positive return, with a Sharpe ratio as high as 5.01.[14]

Panel A of Table 4 compares portfolio performance before and after the inclusion of technical indicators. The *diff* column reports the changes in investment performance resulting from the addition of these characteristics. Overall, the impact of incorporating technical indicators appears to be model-specific, with only a few models exhibiting improved performance. This outcome is reasonable, as our delta-neutral portfolios are

TABLE 3 | Comparison of different hedging strategies.

	Annual return			Sharpe ratio			Max drawdown		
	Spot	Futures	Diff	Spot	Futures	Diff	Spot	Futures	Diff
Lasso	18.06	56.17	**+38.11**	2.63	7.77	+5.14	−9.21	**−2.17**	+7.04
Ridge	16.39	24.21	+7.82	2.08	2.96	+0.88	−12.06	−13.24	−1.18
ENet	18.52	50.00	+31.48	2.72	7.48	+4.77	−9.52	−3.66	+5.86
PLS	22.34	38.88	+16.54	3.16	5.71	+2.56	−4.67	−10.51	−5.85
L-En	21.71	52.04	+30.33	3.20	8.00	+4.81	−9.40	−2.26	**+7.15**
SVR	26.24	43.65	+17.41	3.75	7.58	+3.83	−8.90	−3.96	+4.95
RF	**33.09**	**69.53**	+36.44	**4.18**	8.81	+4.63	−6.35	−4.27	+2.08
GBDT	28.51	64.84	+36.33	3.59	7.95	+4.36	−6.50	−5.14	+1.36
XGBoost	27.70	61.39	+33.68	3.52	7.94	+4.42	−5.26	−3.84	+1.41
DFN	26.33	33.66	+7.32	3.77	5.16	+1.38	−5.57	−3.64	+1.93
N-En	30.24	64.50	+34.26	3.91	**9.09**	**+5.18**	−7.04	−3.92	+3.12
IPCA	2.37	7.28	+4.90	1.06	2.97	+1.91	**−2.26**	−2.57	−0.31

Note: This table compares the investment performance of two types of hedging strategies. The table reports the annualized returns, Sharpe ratio, and maximum drawdown of long-short portfolios based on different machine learning models. *Spot* and *futures* represent spot hedging and futures hedging, respectively. *Diff* shows the investment performance of futures hedging minus that of spot hedging. Highest value in each column is indicated in bold. Numbers about annual return and maximum drawdown are expressed as percentages.

constructed on a daily basis, with most of the underlying asset's price variation effectively hedged through frequent dynamic rebalancing. Consequently, although technical indicators may demonstrate strong predictive power for price movements in single-factor tests, they may offer limited incremental value for portfolio returns in a hedged framework, and may even introduce noise that complicates pattern recognition. Among all models, XGBoost achieves the highest annualized return, while RF delivers the highest Sharpe ratio, further underscoring the robustness of nonlinear models in handling large-scale, high-dimensional data.

Realized measures constructed based on high-frequency data have been shown to exhibit significant and robust relationship with delta-hedged option returns (Cao et al. 2023). As we shall see in Subsection 4.1, high-frequency factor, such as intraday volatility, is one of the most prominent predictors in explaining returns predictability. This evidence inspires us to incorporate realized measures based on high-frequency intraday data and examine their marginal contributions to the investment performance. Specifically, we include realized volatility proposed by Andersen et al. (2003), realized semi-variances to capture upwards and downward movements (Barndorff-Nielsen et al. 2008) and parametric measures of conditional asymmetry to approximate realized skewness (Feunou et al. 2016).

Panel B of Table 4 presents a comparison of portfolio performance before and after incorporating high-frequency intraday realized measures. The results show that almost all nonlinear machine learning models exhibit improvements in both Sharpe ratio and maximum drawdown after integrating additional high-frequency factors, while the enhancements for linear models in constructing profitable portfolios are less pronounced. Specifically, the Sharpe ratio of the nonlinear ensemble model increases from 9.09 to 9.62, while the maximum drawdown declines from −3.92% to −2.30%. These results not only suggest that high-frequency factors can help to enhance model performance by providing more accurate measurements of volatility and skewness, but also demonstrate the advantages of nonlinear models in capturing the information embedded in high-frequency and high-dimensional factors.

3.4 | Source of Predictability

Due to investors' limited information-processing capacity and the substantial costs associated with acquiring accurate forecasting models, some investors can only process a subset of publicly available information (Hong and Stein 1999; Hirshleifer and Teoh 2003), which leads to a gradual diffusion of information across the market and contributes to return predictability. Moreover, limits to arbitrage prevent some investors from correcting mispricings in a timely manner (Pontiff 2006). As a result, new informative signals are only partially incorporated into asset prices, giving rise to predictable return patterns. Motivated by these considerations, we examine whether informational frictions and option mispricing help explain the superior performance of machine learning models in the Chinese options market.[15]

Following Atilgan et al. (2020), we construct an arbitrage index based on multiple variables rather than relying on a single proxy. To build the arbitrage index, we incorporate five characteristics that capture different dimensions of limits to arbitrage. First, we include option trading volume, open interest, and bucket-level illiquidity, following the approach of Bali et al. (2023). Second, we adopt the Amihud (2002) illiquidity measure to capture transaction costs. Third, we include idiosyncratic volatility as a widely used proxy for the arbitrage costs of the underlying asset (Cao and Han 2016; Boulatov et al. 2022).

TABLE 4 | Influence of including additional factors.

	Annual return			Sharpe ratio			Max drawdown		
	With	Without	Diff	With	Without	Diff	With	Without	Diff
Panel A: Characteristics of underlying asset									
Lasso	53.56	56.17	−2.61	7.19	7.77	−0.58	−5.60	**−2.17**	−3.43
Ridge	8.88	24.21	−15.33	1.09	2.96	−1.86	−56.66	−13.24	−43.43
ENet	42.39	50.00	−7.61	5.78	7.48	−1.70	−11.63	−3.66	−7.98
PLS	31.02	38.88	−7.86	4.00	5.71	−1.72	−12.37	−10.51	−1.86
L-En	45.68	52.04	−6.36	6.37	8.00	−1.63	−10.14	−2.26	−7.89
SVR	41.83	43.65	−1.82	7.82	7.58	+0.24	−4.38	−3.96	−0.43
RF	70.09	**69.53**	+0.56	**9.36**	8.81	+0.55	−4.10	−4.27	**+0.17**
GBDT	64.56	64.84	−0.28	7.20	7.95	−0.76	−5.71	−5.14	−0.57
XGBoost	**72.75**	61.39	**+11.36**	9.28	7.94	**+1.34**	−4.13	−3.84	−0.29
DFN	16.15	33.66	−17.50	3.45	5.16	−1.71	−5.16	−3.64	−1.51
N-En	65.85	64.50	+1.35	9.11	**9.09**	+0.02	**−3.80**	−3.92	+0.12
Panel B: High-frequency intraday measures									
Lasso	51.25	56.17	−4.93	7.14	7.77	−0.63	−4.09	**−2.17**	−1.92
Ridge	25.12	24.21	+0.91	3.12	2.96	+0.17	−12.63	−13.24	+0.61
ENet	47.23	50.00	−2.77	6.51	7.48	−0.98	−7.72	−3.66	−4.06
PLS	35.98	38.88	−2.90	5.77	5.71	+0.05	−4.31	−10.51	**+6.20**
L-En	48.39	52.04	−3.65	6.96	8.00	−1.04	−5.25	−2.26	−2.99
SVR	44.18	43.65	+0.53	7.70	7.58	+0.11	−2.83	−3.96	+1.12
RF	**65.21**	**69.53**	−4.32	8.80	8.81	−0.01	−4.26	−4.27	+0.01
GBDT	57.97	64.84	−6.87	8.09	7.95	+0.14	−4.03	−5.14	+1.11
XGBoost	60.33	61.39	−1.05	8.67	7.94	+0.73	−2.90	−3.84	+0.95
DFN	43.44	33.66	**+9.79**	6.44	5.16	**+1.28**	−3.13	−3.64	+0.51
N-En	64.92	64.50	+0.42	**9.62**	**9.09**	+0.53	**−2.30**	−3.92	+1.62

Note: This table compares the investment performance with or without additional factors. Panel A considers the characteristics of underlying asset, while panel B is about high-frequency intraday measures. The table reports the annualized returns, Sharpe ratio, and maximum drawdown of long short portfolios based on different machine learning models. *with* means additional factors have been included, while *without* means it has not. Column *diff* reports the investment performance with additional factors minus that without. Highest value in each column is indicated in bold. Numbers about annual return and maximum drawdown are expressed as a percentage.

To ensure interpretability, we modify the sorting procedure to ensure that higher variable values correspond to tighter limits to arbitrage. Each option is assigned a score based on its decile rank for each variable. The arbitrage index is defined as the sum of these scores, with higher values indicating tighter limits to arbitrage.

Using the arbitrage index, we first examine how informational frictions affect the predictability of option returns. Specifically, we conduct bivariate portfolio sorts to analyze how return predictability vary across different levels of informational frictions. Options are first sorted into tertiles based on the arbitrage index, and then further sorted into quintiles according to the return forecasts generated by the nonlinear ensemble model within each tertile. Panel A of Table 5 compares the differences in return predictability across subsamples with high and low levels of informational frictions. In the group with low arbitrage costs, the return spread between the highest and lowest forecast quintiles is close to zero, indicating weak return predictability. By contrast, in the group with high arbitrage costs, the average return spread between the top and bottom forecast quintiles reaches 0.051% per day with a t-statistic of 2.92. The difference-in-differences return spread between the high and low arbitrage cost groups is 0.050% per day, which is both economically and statistically significant. These findings suggest that informational frictions represent an important source of the superior performance of nonlinear machine learning models in predicting option returns.

Because machine learning models are able to identify misvaluation in the options market, greater return predictability is naturally observed among mispriced options. Using the similar approach, we estimate the level of mispricing per option contract using a composite mispricing index. Specifically, we include three components: the pricing error between theoretical and observed option prices (Boulatov et al. 2022), the variance spread (Goyal and Saretto 2009; Carr and Wu 2009), and the absolute return prediction of the nonlinear ensemble model (Bali et al. 2023). Similarly, each option receives a score based on its decile rank for each of the three constitutional variables. The mispricing index is defined as the sum of these scores, with higher values indicating more severe mispricing.

TABLE 5 | Bivariate portfolios of option index and expected returns.

	Low pred.	2	3	4	High pred.	H–L	t(H–L)
Panel A: Arbitrage index							
Low	−0.028	−0.025	−0.035	−0.033	−0.028	0.001	(0.06)
High	−0.179***	−0.121***	−0.111***	−0.124***	−0.128***	0.051***	(2.92)
H–L	−0.151***	−0.096***	−0.076***	−0.091***	−0.100***	0.050***	(2.69)
t(H–L)	(−9.22)	(−7.57)	(−5.47)	(−6.54)	(−6.69)		
Panel B: Mispricing index							
Low	0.003	−0.001	−0.002	0.027	0.070***	0.066***	(7.28)
High	−0.261***	−0.196***	−0.174***	−0.134***	−0.132***	0.129***	(8.64)
H–L	−0.264***	−0.195***	−0.173***	−0.160***	−0.201***	0.063***	(3.97)
t(H–L)	(−17.58)	(−14.85)	(−13.95)	(−12.70)	(−15.43)		

Note: The table shows realized returns to delta-neutral portfolios first sorted by an option index, either about arbitrage cost (panel A) or about the mispricing (panel B) and then by the return predictions made by nonlinear ensemble model. Option open interest, trading volume, bucket-level illiquidity, Amihud (2002) illiquidity and idiosyncratic volatility are taken for constructing the arbitrage score. Since the level of open interest and trading volume are inversely related to informational frictions, these characteristics are inversely sorted in this indices' construction. The pricing error between theoretical and observed option prices, variance spread and the absolute return prediction of the nonlinear ensemble are used for constructing the mispricing score. H–L is the time-series average of the differences in returns between the High and Low subsamples. The *t*-statistics are adjusted by the Newey and West (1987) method and reported in parentheses. Significance at the 1%, 5%, and 10% level is indicated by ***, **, *, respectively.

Panel B of Table 5 compares the differences in return predictability across subsamples with high and low levels of option mispricing. In the group with low option mispricing, the return spread between the highest and lowest forecast quintiles is 0.066% per day. By contrast, in the group with high option mispricing, the return spread is 0.129% per day, resulting in an economically and statistically significant difference-in-differences return spread of 0.063% per day. These results support the intuition that machine learning models can effectively capture mispricing in the options market and suggest that option mispricing is an important source of option return predictability.

3.5 | Transaction Costs

Previous sections assume that assets were traded at mid-price, disregarding the potential impact of bid-ask spreads. From a more realistic perspective, every transaction entails implicit costs, and existing research has established that costs in the options market can be substantial (Ofek et al. 2004). Therefore, this paper further investigates the influence of transaction costs on machine learning-based multi-factor quantamental investment. Four different scenarios are considered: 25%, 50%, and 100% of the bid-ask spread, as well as a scenario that fully accounts for the bid-ask spread in addition to option margin requirements.

Table 6 reports the investment performance under varying levels of transaction costs. The results indicate that even when accounting for 25% or 50% of the bid-ask spread, machine learning-driven multi-factor strategies still achieve positive annual returns and Sharpe ratios. However, as transaction costs increase, the performance of the constructed delta-neutral portfolios deteriorates significantly. At a 100% bid-ask spread level, portfolios based on Ridge regression and DFN fail to pass the Newey-West significance tests. When margin requirements for option trading are further incorporated, the Sharpe ratio of the best-performing nonlinear model declines to 2.48, while its linear counterpart drops to 2.23. These findings highlight the considerable impact of trading costs on the profitability of long-short strategies generated by machine learning algorithms. Only models with sufficient robustness are able to sustain economically meaningful excess returns under realistic trading frictions. Given that the inclusion of bid-ask spreads and margin requirements closely reflects actual market conditions, our results suggest that carefully selected machine learning-based multi-factor strategies can still generate positive risk-adjusted returns, even after fully accounting for transaction costs.

3.6 | Generalization Ability

The Chinese options market is undergoing rapid development, with new contracts being introduced continuously. Accurately pricing newly issued options has become a topic of considerable interest. Prior research has demonstrated that machine learning algorithms possess strong generalization capabilities (Bishop and Nasrabadi 2006), allowing them to perform well on out-of-sample data. This paper aims to explore and exploit this generalization ability to predict the returns of newly launched options. Due to the lack of historical data for new products, models must learn meaningful relationships from existing long-standing contracts. Specifically, we train our models using factor data from options issued before 2020 and evaluate their predictive performance on contracts introduced after July 2022. Because the training set contains the earliest and most established products in the Chinese options market, it provides sufficient information for the model to learn generalizable patterns linking characteristics to returns.

Table 7 reports the investment performance of machine learning models on newly issued option contracts under varying levels of transaction costs. The results suggest that these models effectively extract useful patterns during training and exhibit strong out-of-sample generalization ability. Notably, machine

TABLE 6 | Investment performance with different transaction costs.

	Lasso	Ridge	ENet	PLS	L-En	SVR	RF	GBDT	XGBoost	DFN	N-En
Annual Return											
0%	56.17	24.21	50.00	38.88	52.04	43.65	**69.53**	64.84	61.39	33.66	64.50
25%	43.78	15.64	35.74	28.32	42.77	37.03	50.08	47.19	49.82	30.24	**53.14**
50%	33.38	7.71	30.99	18.89	32.94	22.35	**43.04**	27.30	33.77	18.43	39.74
100%	13.87	−6.19	12.46	2.04	13.88	3.42	**20.22**	8.84	0.98	−3.93	15.32
Margin	11.72	−7.82	8.45	−2.52	12.37	1.46	**13.33**	5.84	9.32	−1.95	13.29
Sharpe Ratio											
0%	7.77	2.96	7.48	5.71	8.00	7.58	8.81	7.95	7.94	5.16	**9.09**
25%	6.79	2.04	5.62	4.42	6.70	6.35	7.39	6.79	7.65	4.71	**7.95**
50%	5.12	1.05	5.14	3.09	5.42	4.53	**6.33**	4.18	5.30	3.84	5.95
100%	2.31	−0.82	2.15	0.38	2.45	0.79	**3.17**	1.50	0.22	−0.68	2.83
Margin	2.03	−1.07	1.49	−0.43	2.23	0.35	2.17	1.02	1.82	−0.29	**2.48**
t-statistics											
0%	20.14	8.54	19.94	15.21	21.95	20.63	23.75	22.20	21.29	**39.48**	25.01
25%	17.81	5.96	15.53	11.87	17.68	17.80	20.32	18.60	20.87	13.87	**21.42**
50%	13.42	3.13	13.91	8.32	14.35	12.64	**17.21**	11.76	14.98	10.77	16.24
100%	5.95	−2.32	5.59	1.05	6.38	2.21	**8.65**	3.97	0.49	−2.04	7.56
Margin	5.20	−3.02	3.97	−1.08	5.78	0.98	5.67	2.75	4.67	−0.80	**6.54**

Note: This table reports the investment performance of machine learning under different transaction costs. Long-short portfolios are constructed and *Margin* considers both 100% bid-ask spread and margin requirement. Investment performance are evaluated by annualized return, Sharpe ratio and *t*-statistics adjusted by Newey and West (1987) method. Highest value in each row is indicated in bold. Numbers about annual return are expressed as a percentage.

TABLE 7 | Evaluation of machine learning generalization ability.

	Lasso	Ridge	ENet	PLS	L-En	SVR	RF	GBDT	XGBoost	DFN	N-En
Annual return											
0%	124.45	68.87	111.62	60.38	110.27	86.99	**176.20**	151.44	115.79	99.44	146.54
25%	107.36	56.63	94.08	48.32	105.27	53.27	**138.56**	127.95	117.25	54.60	127.30
50%	87.58	46.66	82.80	36.58	92.11	53.44	**115.88**	100.68	98.61	39.46	112.90
100%	59.96	27.48	46.58	15.79	48.19	30.79	**87.30**	55.98	64.68	31.34	71.28
Margin	59.20	24.91	58.02	13.96	49.43	30.32	**80.15**	55.88	69.97	11.63	65.81
Sharpe ratio											
0%	8.70	7.48	8.36	6.44	7.96	10.79	13.48	13.37	11.74	8.69	**13.67**
25%	7.84	6.82	7.43	5.41	7.75	8.61	**12.35**	11.03	11.67	6.45	12.13
50%	6.82	5.80	6.68	4.29	7.07	7.86	10.36	10.33	10.20	4.75	**10.59**
100%	5.11	3.71	4.25	2.04	4.33	4.95	8.10	6.40	7.53	4.61	**8.46**
Margin	5.05	3.37	4.88	1.84	4.92	4.76	**8.02**	6.65	7.71	1.85	7.89
t-statistics											
0%	10.65	9.08	10.29	8.19	8.86	12.33	14.91	14.71	13.08	9.82	**15.58**
25%	9.59	8.39	9.26	6.88	9.65	9.89	13.46	12.86	12.65	7.62	**13.57**
50%	8.39	7.07	8.30	5.47	8.80	9.06	11.43	11.51	11.23	5.71	**11.98**
100%	6.31	4.53	5.28	2.61	4.83	5.76	8.95	7.30	8.18	4.99	**9.56**
Margin	6.24	4.10	6.06	2.33	5.55	5.49	**9.15**	7.74	8.49	2.29	8.83

Note: This table reports investment performance of machine learning on newly issued option products. Models are trained using factor-return data of the long-standing option contracts. Delta-neutral portfolio is hedged with futures and evaluated under different levels of transaction costs. Performance is measured by annual return, Sharpe ratio, and *t*-statistics adjusted by Newey and West (1987) method. Highest value in each row is indicated in bold. Numbers about annual return are expressed as a percentage.

learning models achieve better investment performance on newly launched options compared to long-standing products. This superior performance is partly attributable to the relatively larger pricing inefficiencies and greater arbitrage opportunities commonly observed in new contracts. In addition, newly issued options are typically linked to small- and mid-cap stock indices or ETFs, which tend to exhibit higher growth potential and greater price volatility. To enhance portfolio liquidity and mitigate slippage risk, each contract is weighted by its dollar open interest at the time of investment. While newly issued products offer higher risk-adjusted returns, the overall profitability in the on-exchange options market is still largely driven by a limited number of highly liquid products with greater market participation and more stable return profiles. In summary, these findings provide strong evidence for the generalization capacity of machine learning models in predicting the returns of newly introduced options.

4 | Factors Analysis

Given its extensive number, it is critical to identify truly important predictors from factor zoo (Cochrane 2011). This paper next investigates whether certain predictors hold greater importance than others and analyzes their impact on model predictions.

4.1 | Factors Evaluation

This section analyzes factor importance under different hedging strategies using the SHAP method.[16] Given the variation in predictive accuracy and factor relevance across models, we weight SHAP values by the corresponding model's RankIC scores[17] to enhance interpretability and reduce bias in the assessment. To facilitate comparison, factor importance values are normalized such that the average importance of the most influential predictor equals one across all scenarios. Figure 2 presents the ranking of factors based on their importance in the testing set, visualized through boxplots for the top 15 factors under different trading strategies. In the figure, whiskers denote the 1st and 99th percentiles, while blue dots represent the mean importance values. As noted in Section 2.4, all option-related factors are classified into six major categories. The corresponding category abbreviation is shown in parentheses following each factor name.

Panel A of Figure 2 displays the distribution of predictor importance under the futures hedging strategy. The most influential factor for return prediction is implied volatility, followed by near-the-money call-minus-put implied volatility (*civpiv*), the sensitivity of option price to time decay (*theta*), and a modified illiquidity measure based on zero-return frequency (*pfht*). Among the six major factor categories, risk-related characteristics are the most dominant, accounting for nearly half of the top predictors. Factors related to illiquidity and informed trading also appear prominently, with four and three representatives, respectively. Although only one factor pertains to contract-specific information, it ranks as the single most influential variable in the model.

Similarly, we assess predictor importance under the spot-hedging strategy and visualize the top 15 most impactful characteristics using boxplots in Panel B of Figure 2. Implied volatility remains the most significant factor in predicting delta-neutral returns, followed by near-month implied variance asymmetry (*ivarud30*), the put-call ratio (*pcratio*), and at-the-money implied volatility (*atm_iv*). Among the major factor categories, risk-related characteristics still dominate, accounting for more than half of the selected predictors. Informed trading contributes three influential factors, while contract information, informational frictions, and illiquidity each contributes one key predictor.

4.1.1 | Comparison of Different Hedging Strategies

By comparing the importance distributions under different hedging strategies, we identify several common factors that consistently exert significant influence on delta-neutral returns. First, implied volatility (IV) emerges as the most critical variable for predicting option price dynamics. It not only reflects the market's expectations regarding future volatility but also signals potential trading opportunities. When IV is overestimated, shorting high-IV options tends to yield positive returns, provided that delta risk is properly managed. Second, the prominence of embedded leverage (*embedlev*) across both hedging strategies is largely attributable to the fact that most listed financial options in China are written on indices and ETFs. *Embedlev*, which measures the magnification of option returns relative to the underlying asset, is generally higher for index options than for individual equities. Its predictive importance aligns with the findings of Frazzini and Pedersen (2022), who highlight that embedded leverage can amplify both risk and return. Third, the daily construction of delta-neutral portfolios enhances the relevance of high-frequency features. For instance, intraday volatility, calculated from 5-min return intervals (Cao et al. 2023), consistently ranks among the top predictors regardless of the hedging approach. This demonstrates that trading signals derived from high-frequency data can significantly improve model performance. Additional common factors include implied volatility asymmetry (*ivarud30*), put-call ratio (*pcratio*), risk-neutral kurtosis (*rnk91-scaled*), as well as option gamma and theta. These characteristics effectively capture cross-sectional variations in option returns across different portfolio construction methods, underscoring their robustness and generalizability. Collectively, these results suggest the existence of core driving forces behind delta-neutral returns that operate independently of specific hedging strategies.

Factor importance varies notably with the choice of hedging strategy. The spot hedging strategy emphasizes characteristics reflecting short-selling constraints, whereas the futures hedging strategy highlights factors associated with asset liquidity. On the one hand, restrictions on short selling in the spot market elevate the demand for and premium of put options (Evans et al. 2009). Consequently, factors related to put-call parity deviations (*pcpv30* and *vs_level*), put-call demand imbalances (*pcratio*), and implied volatility asymmetry (*ivarud30*) play a crucial role in return prediction under the spot hedging strategy. However, their relative importance declines under futures hedging, with only a subset remaining among the top 15 most influential predictors. These results provide further evidence that

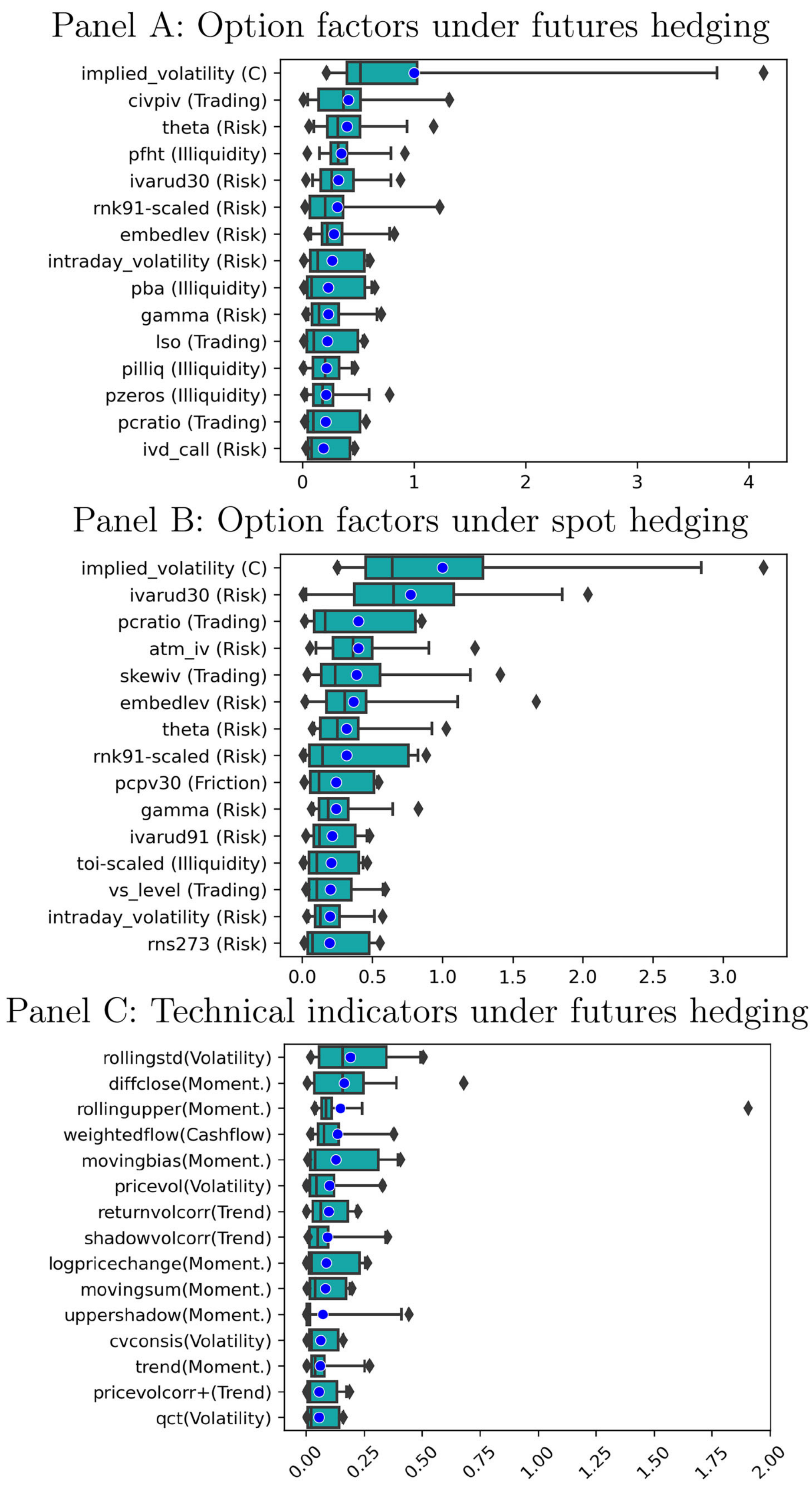

FIGURE 2 | Importance dispersion of the 15 selected factors. Panel A: Option factors under futures hedging. Panel B: Option factors under spot hedging. Panel C: Technical indicators under futures hedging. *Note:* This figure displays the distributions of factor importance for the 15 most influential predictors. For comparability, the importance of each characteristic is normalized by the average SHAP magnitude of implied volatility within the corresponding scenario. Panel C excludes option characteristics and visualizes results solely for technical indicators.

short-sale constraints constitute a market imperfection and serve as a key determinant of option returns (Ramachandran and Tayal 2021). On the other hand, under futures hedging, liquidity-related characteristics, such as proportional bid-ask spread (*pba*), percentage illiquidity (*pilliq*), and the fraction of zero return days (*pzeros* and *pfht*), exhibit stronger predictive power. By contrast, only a few illiquidity-related variables rank among the top predictors in the spot hedging context. This divergence marks a key difference in group-level factor effectiveness. As illustrated in Figure 3, illiquidity measures are more impactful under futures hedging, whereas informational frictions are more influential under spot hedging.

4.1.2 | Comparison to the US Options Market

As introduced in Section 1, the Chinese options market distinguish from that of the U.S. in many aspects, which could lead to substantial differences in factor importance. We conduct a systematic comparison with the results reported in Bali et al. (2023)

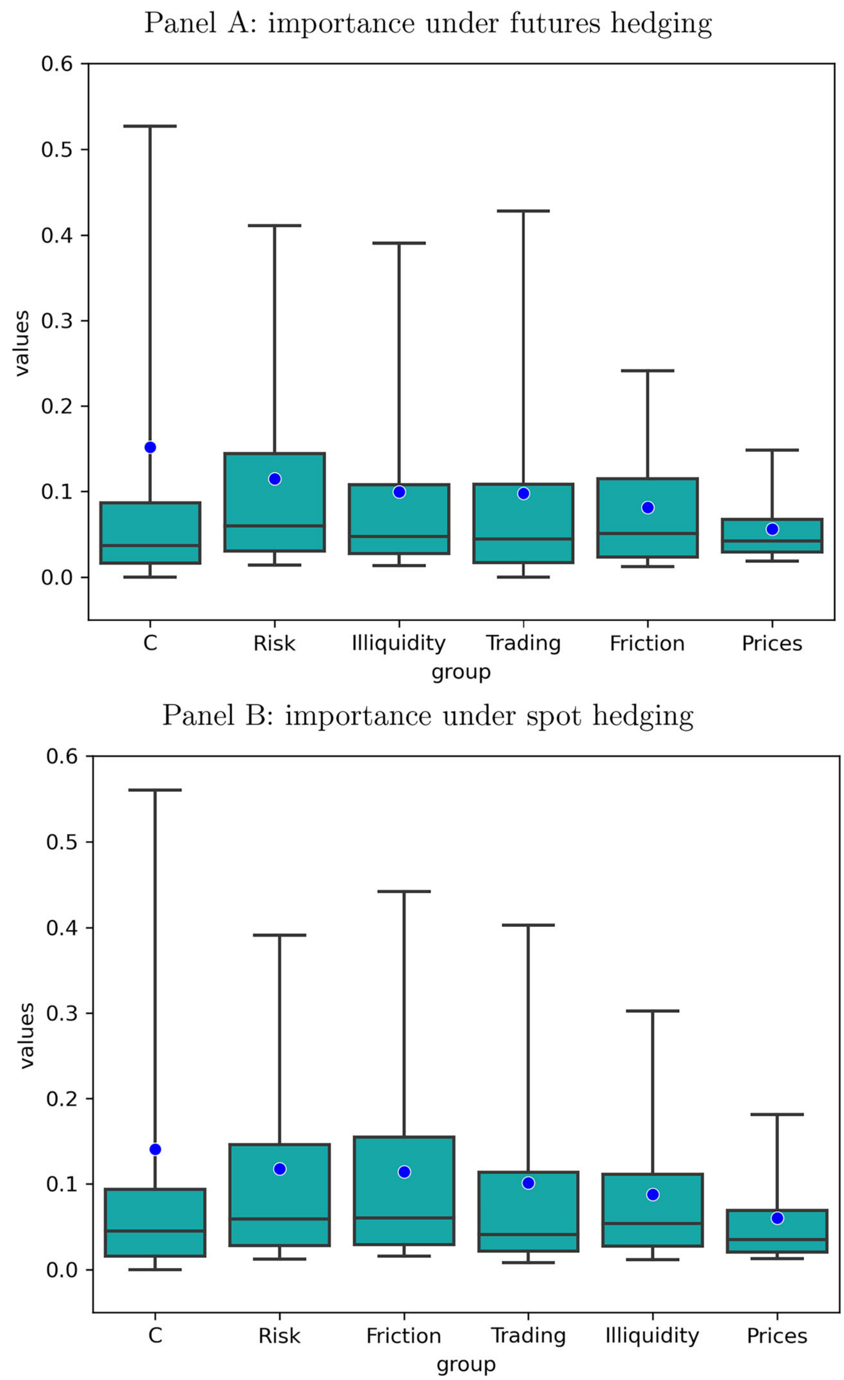

FIGURE 3 | Feature importance. Panel A: importance under futures hedging. Panel B: importance under spot hedging. *Note:* This figure illustrates the distributions of factor importance across the six major groups of option characteristics. Each characteristic's importance is normalized by the average SHAP magnitude of implied volatility. The whiskers denote the 5th and 95th percentiles, and the blue dots indicate the mean importance within each group.

for the US options market, using a consistent spot hedging strategy. Implied volatility, at-the-money implied volatility and *theta* are found to be important in both countries. However, factors related to put-call parity deviations and jump risk are more significant in China, while those associated with retail trading are less important. This pattern reflects the unique characteristics of the Chinese options market, particularly in terms of short-selling constraints, asset liquidity, and investor structure.

Firstly, compared to the US market, short-sale constraints in the Chinese spot market make factors related to imbalanced option demand and put-call parity violations more influential in predicting delta-neutral returns. On the one hand, short-sale constraints affect option demand: overpriced stocks tend to attract greater short-selling pressure, which in turn increases demand for put options. As a result, the put-call ratio (*pcratio*) emerges as a particularly strong predictor of delta-neutral returns in the Chinese options market. On the other hand, short-sale constraints also influence hedging costs. The difficulty of borrowing shares in the equity lending market is reflected in option prices, often making put options more expensive and leading to violations of put-call parity. Consequently, indicators of put-call parity deviation, such as *pcpv30* and *vs_level*, play a more prominent role in forecasting option returns in China. However, none of these variables rank among the top 10 most influential characteristics in the US options market, as reported by Bali et al. (2023). These findings are consistent with Ramachandran and Tayal (2021), and highlight how short-sale constraints in the underlying equity market can shape option return predictability through a multi-factor perspective.

Secondly, trading restrictions in the Chinese financial market also hinder asset liquidity, making factors related to jump risk more pivotal in predicting delta-neutral returns. Proxies for jump risk, such as the implied volatility smirk measure (*skewiv*), risk-neutral skewness (*rns273*), risk-neutral kurtosis (*rnk91-scaled*), and option gamma, play a more significant role in forecasting delta-neutral returns in the Chinese options market. By contrast, in the US options market examined by Bali et al. (2023), these jump risk proxies are ranked substantially lower in factor importance. These findings are consistent with prior literature suggesting that illiquid stocks and options tend to exhibit skewed, fat-tailed return distributions, which are closely associated with notable jump risk premia (Christoffersen et al. 2018; Zhan et al. 2022). Besides, implied variance asymmetry (*ivarud30*), which measures the relative strength of extreme upside versus downside returns under the risk-neutral measure, is also highly important in the Chinese options market, ranking just behind implied volatility under the spot hedging strategy. This finding aligns with Huang and Li (2019), who demonstrate that implied variance asymmetry is particularly predictive for stocks subject to severe limits to arbitrage.

Thirdly, institutional investors are the primary participants in the Chinese options market, which results in the limited significance of factors associated with retail trading. The disposition effect is one of the most well-documented behavioral biases among individual investors (Barberis and Xiong 2009). Bergsma et al. (2020) propose a novel proxy, *ocgo*, to measure disposition effect and demonstrate its significant relationship with US equity option returns. Besides, the Chinese stock market is well known for the dominant presence of retail investors (Leippold et al. 2022). Sentiment-based factors, such as turnover, play a crucial role in explaining the cross-section of returns in the Chinese stock market (Liu et al. 2019; Maasoumi et al. 2024). However, both *ocgo* and turnover are relatively less important in predicting delta-neutral returns in the Chinese options market, ranking 57th and 87th out of 119 characteristics, respectively. This finding reflects the institutional dominance in the Chinese options market and suggests that factors capturing retail sentiment or behavioral biases play a limited role in shaping option return predictability.

4.1.3 | Evaluation of Technical Indicators

This paper also analyzes the importance of technical indicators. During model training and evaluation, both option characteristics and technical indicators are used as inputs. However, Panel C of Figure 2 focuses exclusively on technical indicators and presents boxplots for the 15 most influential ones, as the SHAP values of technical indicators and option characteristics differ substantially in scale and magnitude.

Panel C in Figure 2 reveals that, for delta-neutral portfolios under futures hedging strategy, the most impactful technical indicator is *rollingstd*, which captures the standard deviation of closing prices over the past few periods. Closely following are *diffclose*, which represents the change in closing price over the given time interval, and *rollingupper*, which reflects the relationship between the initial, closing, and historical peak prices within the given time window. The constructed technical indicators are categorized into six major groups. Among the selected 15 indicators, characteristics about momentum are the most prevalent with a total number of 7, followed by 4 volatility-related, 3 trend-related characteristics and one concerning cashflow. Compared with technical indicators, option characteristics exhibit significantly greater predictive power. Specifically, the most important option characteristic, that is, implied volatility, has an average importance several times higher than that of any technical indicator.[18] This result corroborates the earlier findings in Section 3.3, wherein the inclusion of technical indicators failed to yield statistically significant improvements in annual returns or Sharpe ratios for machine learning-driven delta-neutral portfolios.

4.2 | Marginal Association

To shed light on how predictors influence predictions, this paper visualizes the model-implied marginal effects of selected characteristics on expected returns in Figure 4. Following Gu et al. (2020), we normalize characteristics to the $[-1, 1]$ interval and holds all other variables fixed at their median value. Based on the analysis in Section 4.1, we select several key characteristics for illustration, including implied volatility (Contract), intraday volatility (Risk), *lso* (Informed Trading), *pfht* (Illiquidity), and *pcpv30* (Friction).

Firstly, machine learning models are capable of identifying empirical patterns that align with findings in the existing

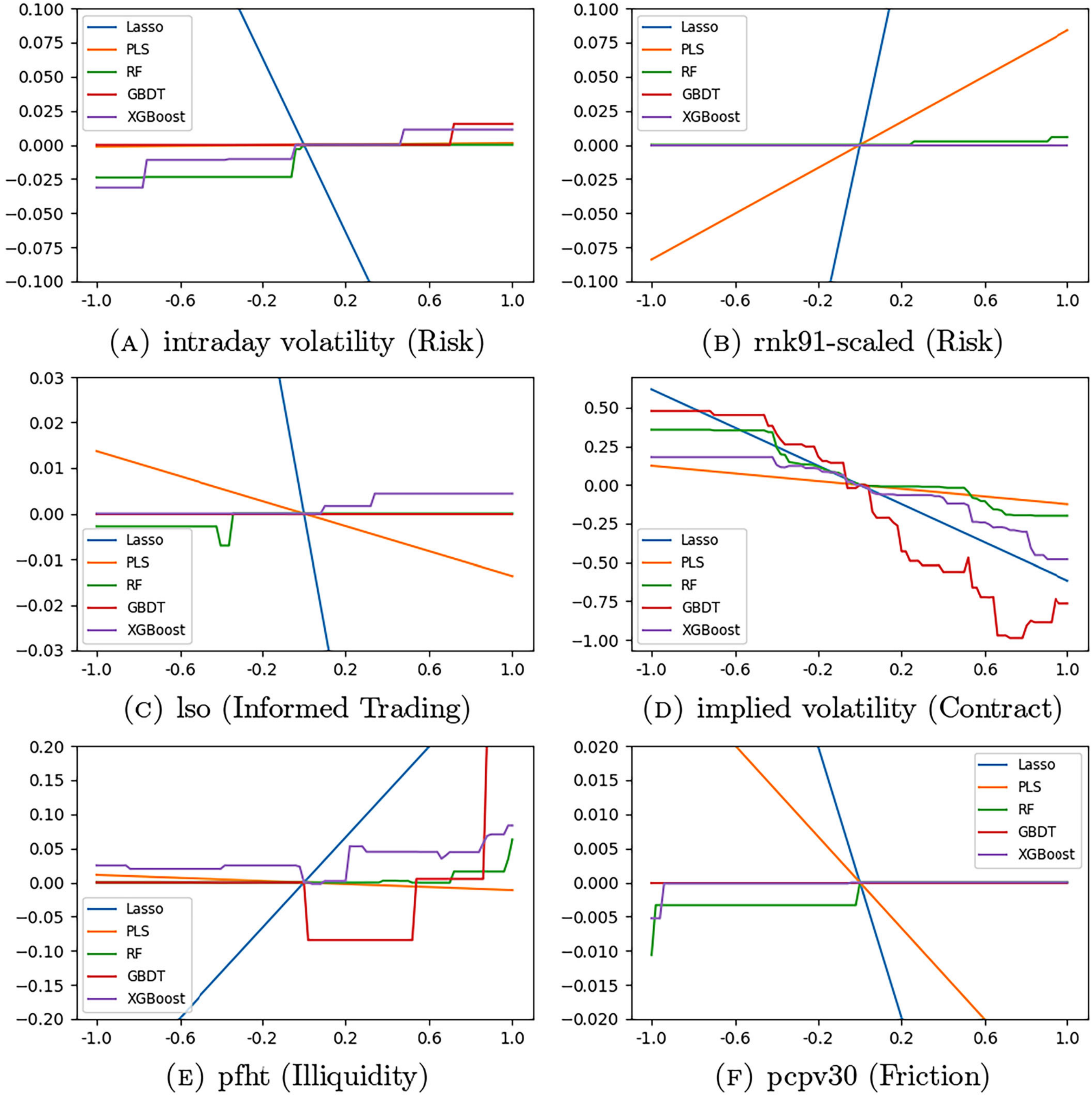

FIGURE 4 | Marginal association between expected returns and characteristics. (A) intraday volatility (Risk). (B) rnk91-scaled (Risk). (C) lso (Informed Trading). (D) implied volatility (Contract). (E) pfht (Illiquidity). (F) pcpv30 (Friction). *Note:* This figure illustrates the sensitivity of expected percentage returns (vertical axis) to individual characteristics. The representative characteristics include intraday volatility, scaled near-month risk-neutral kurtosis (*rnk91-scaled*), the logarithm of stock volume relative to option volume (*lso*), implied volatility, an illiquidity measure based on zero returns (*pfht*), and short-term put-call parity deviations (*pcpv30*). The specific characteristic analyzed is reported in the subfigure captions, with its corresponding group indicated in parenthesis. All other covariates are held fixed at their median values.

literature. Figure 4d illustrates a negative correlation between expected returns and implied volatility (IV). Higher IV is typically associated with higher option prices, implying higher cost of establishing long position. Moreover, IV exhibits mean-reverting property (Jackwerth and Rubinstein 1996; Christensen and Prabhala 1998). In delta-neutral portfolios, initiating a long position when IV is high (low) entails paying a higher (lower) premium and may result in larger losses (gains) as IV reverts to its normal level. Second, nonlinear models are better equipped to capture complex relationships that are difficult for linear models to identify. For the illiquidity characteristic *pfht*, the marginal relationship with expected returns is non-monotonic: returns initially decline and then rise as illustrated in

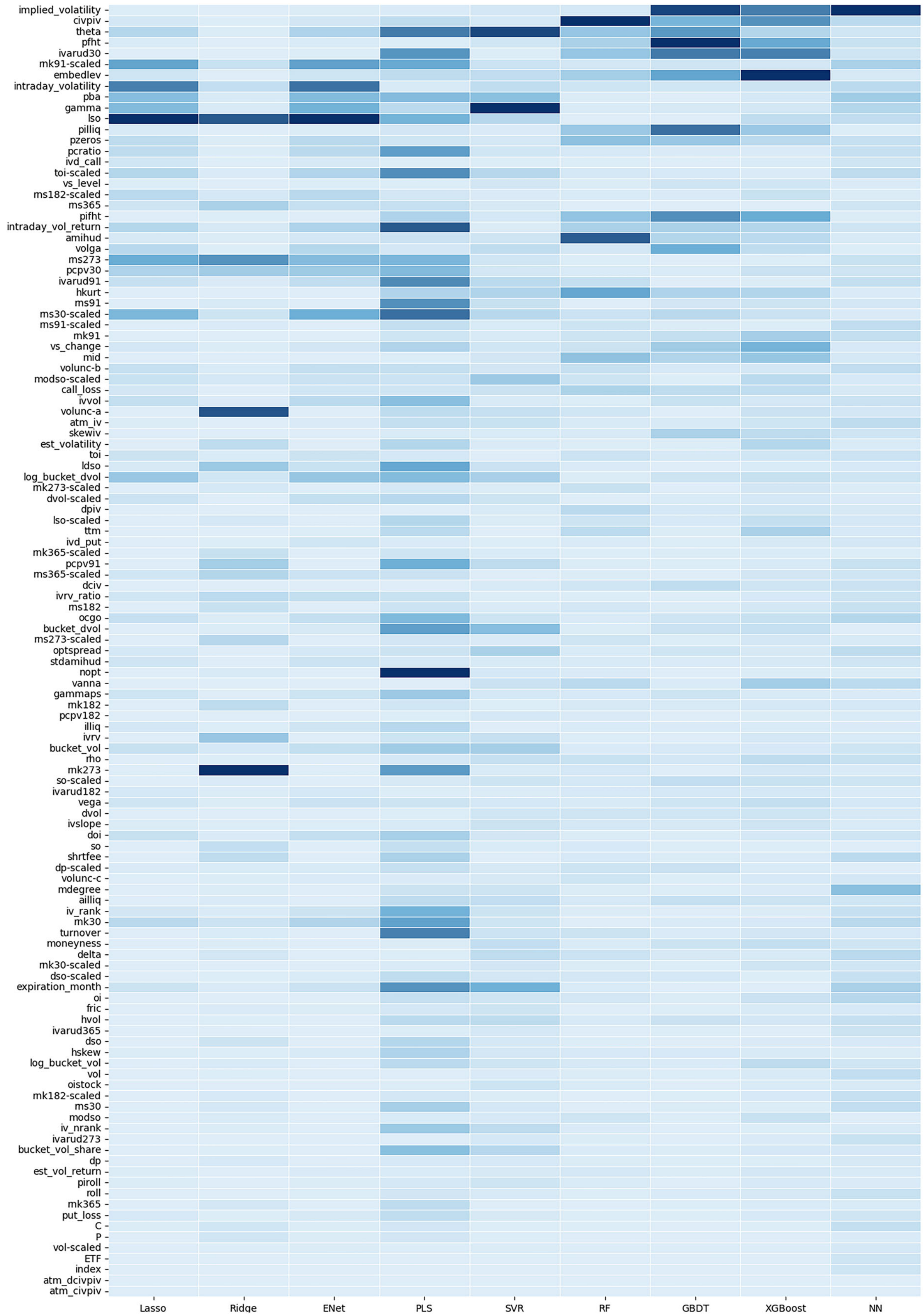

FIGURE 5 | Characteristic importance. *Note:* This table ranks 119 option characteristics based on their overall contribution to model predictions. Characteristics are ordered by the weighted sum of their absolute SHAP values across all models, with the most influential characteristics listed at the top. Each column corresponds to an individual model, and the color gradient within each column ranges from dark blue (most influential) to white (least influential).

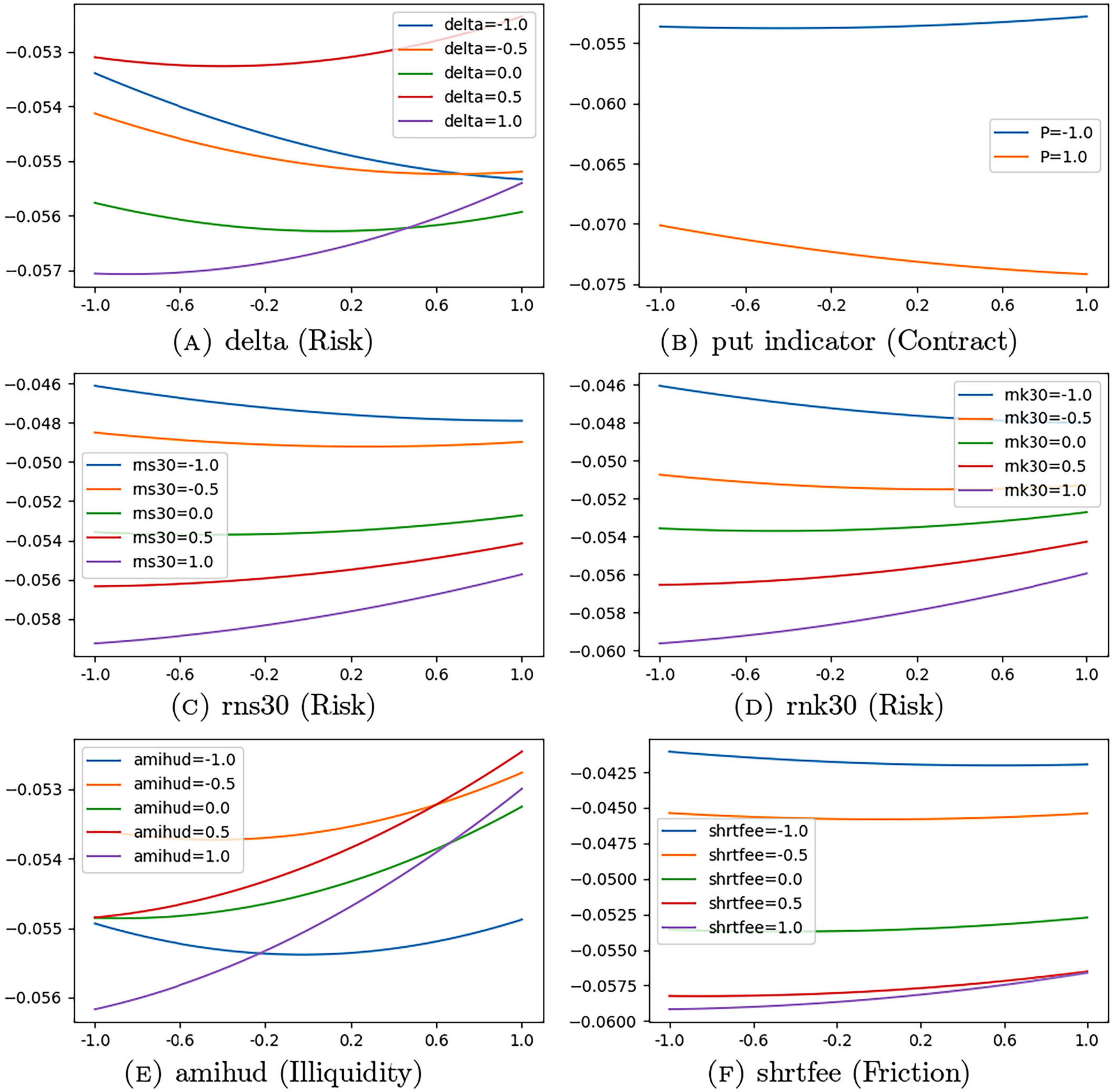

FIGURE 6 | Expected returns and characteristic interactions (N-En). (A) delta (Risk). (B) put indicator (Contract). (C) rns30 (Risk). (D) rnk30 (Risk). (E) amihud (Illiquidity). (F) shrtfee (Friction). *Note:* This figure illustrates the sensitivity of expected percentage returns (vertical axis) to the interaction effects between risk-neutral skewness (*rns30*), risk-neutral kurtosis (*rnk30*), implied volatility, put indicator, *amihud*, and *shrtfee* with scaled demand pressure (dp-scaled) in the N-En model. The put indicator is a binary variable, where +1 represents a put option and −1 otherwise. The specific characteristic analyzed is reported in the subfigure captions, with its corresponding group indicated in parenthesis. All other covariates are held fixed at their median values.

Figure 4e. Nonlinear models successfully detect these "jumps" in the return curve, while linear models struggle to capture such abrupt shifts, evidenced by Lasso and PLS yielding markedly different linear approximations, one steep and the other relatively smooth. Third, marginal association analysis echoes the factor importance evaluation obtained via the SHAP method. Figure 5 reports the characteristics importance for each model, where the color gradient within each column reflects the model-specific importance from least to most (lightest to darkest). Notably, factors highlighted with darkest color by linear models, such as intraday volatility and *rnk91-scaled*, exhibit steeper slopes in Figure 4a,b.[19] By contrast, nonlinear models assign higher importance to characteristics like implied volatility, which span a wider range in Figure 4d. The inability of linear models to capture such complicate relationships can lead to misidentification of important characteristics, partially explaining their inferior performance in forecasting asset returns.

4.3 | Interaction Effect

The favorable performance of tree-based models highlights the advantage of allowing for complex interactions among predictors. However, it is challenging to explore interaction effects due to the overwhelming number of potential functional forms. In this subsection, we present several representative interaction results to illuminate the underlying mechanisms of nonlinear ensemble models. Existing literature has documented the influence of demand pressure on option prices (Bollen and Whaley 2004; Garleanu et al. 2008). For instance, Bollen and Whaley (2004) found that in the US market, buying pressure can reshape the implied volatility function, with index put options being particularly affected. We adopt the same methodology of Cao et al. (2023) to measure demand pressure[20] and investigate its interaction effect with other option characteristics.

Figure 6 depicts how expected returns vary as demand pressures range from −1 to 1, with representative characteristics set at different levels and all other predictors fixed at their median values. In Figure 6a, the interaction effect between expected returns and demand pressure exhibits distinct patterns depending on the level of delta. When delta is relatively high, a positive relationship emerges. However, this relationship turns negative when delta is relatively low. Furthermore, our analysis confirms discovery of Bollen and Whaley (2004) by showing that the impact of demand pressure is stronger when put indicator equals to +1, indicating an index put option type, as shown in Figure 6b. It can be observed that the influence of demand pressure towards expected returns is more pronounced when near-month risk neutral skewness and kurtosis are low, leading to nonparallel variation curves in Figure 6c,d. In addition, characteristics from other major option groups, such as illiquidity (Figure 6e) and friction (Figure 6f), also exhibit nonlinear interaction patterns with demand pressure, further highlighting the complexity of factor interdependencies captured by nonlinear models.

In summary, the interaction effects between option characteristics can deliver a substantial impact on expected returns, underscoring the importance of capturing nonlinear relationships when constructing delta-neutral portfolios. These findings provide a compelling explanation for the superior performance of nonlinear machine learning models in China's unique financial context.[21]

5 | Conclusion

This paper investigates the predictive power of machine learning methods in the Chinese options market. We find that machine learning models outperform the IPCA benchmark by extracting useful information from high-dimensional option characteristics, and nonlinear models more effectively span the efficient frontiers than linear models. By comparison, spot hedging relies more on factors related to put-call parity deviations induced by short-selling constraints, whereas futures hedging places greater emphasis on asset liquidity. Our analysis further reveals that informational frictions and mispricing are key sources of option return predictability. Machine learning models demonstrate strong predictive power on newly issued option products, owing to their superior generalization capabilities. In addition, robustness tests show that machine learning-based quantamental strategies can generate positive risk-adjusted returns even after fully accounting for transaction costs.

We further systematically investigate the factors that drive option return predictability. Compared to the US market, the Chinese options market places greater importance on factors related to put-call parity violations, jump risk, and less importance on those associated with retail trading, reflecting its distinct structural characteristics in terms of short-selling constraints, asset liquidity, and investor composition. We also identify complicate relationship and interaction effects among predictors by efficient interpretability analysis to illuminate option pricing dynamics and inner mechanism of nonlinear models. Overall, this paper demonstrates that machine learning methods can be successfully applied to markets that have completely different features than the US market.

Author Contributions

Yuxiang Huang: conceptualization, methodology, data curation, software, formal analysis, writing – original draft. **Zhuo Wang:** conceptualization, methodology, investigation, formal analysis, supervision, writing – review and editing. **Zhengyan Xiao:** conceptualization, methodology, supervision, writing – original draft.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the finding of this paper are available from the authors upon reasonable request.

Endnotes

[1] The report can be found at https://www.sse.com.cn/aboutus/research/report/.

[2] Calculated by using the average exchange rate (USD/CNY = 7.0774) in 2023.

[3] These statistics are from https://www.cboe.com/us/options/market_statistics/historical_data/.

[4] Statistics are from https://www.sse.com.cn/aboutus/research/report/.

[5] STAR50 ETF is an Exchange Traded Fund capturing China's most promising tech companies.

[6] For example, when constructing risk-neutral kurtosis and skewness, we scale the option prices and strike prices to facilitate comparison between options with different types of underlying assets.

[7] More details about training procedures can be found in Supporting Information S1: Appendix A. The comparative analysis to examine

the influence of different retrain frequency is presented in Supporting Information S1: Appendix G.

[8]Both Information Coefficient (IC) and Rank Information Coefficient (RankIC) are used to measure the correlation between predicted and actual returns. Higher values indicate superior model performance.

[9]According to a research report published by CITIC Securities, the combined daily trading value of on-exchange financial options and commodity options in 2023 reached 97.67 billion yuan, with financial options contributing 54.99 billion yuan, accounting for approximately 56.3%. The average daily open interest for both categories was 13.087 million contracts, with financial options representing 7.614 million contracts, or about 58.2%.

[10]Following Bali et al. (2023), we first divide options into different buckets depending on their time to maturity and standardized moneyness, and then aggregate characteristics from options in the same bucket to obtain bucket-level factors.

[11]These factors include time to maturity, embedded leverage, implied volatility, and four Greek letters describing different risk exposures.

[12]Spot (futures) hedging strategy calculates option characteristics such as implied volatility and delta based on spot (futures) prices and construct delta-neutral portfolio by adjusting the position of spot assets (futures contracts).

[13]For example, differences in implied volatility affects the calculation of standardized moneyness, which leads to options being classified into different buckets and further impacts the bucket-level characteristics.

[14]Detailed methodologies for constructing technical indicators are provided in Supporting Information S1: Appendix E, while the results of the single-factor tests are presented in Supporting Information S1: Appendix F.

[15]We thank the anonymous referee for the suggestion to examine the economic mechanisms underlying the superior performance of machine learning models in the Chinese options market.

[16]SHAP is an efficient model interpretation framework based on cooperative game theory, which quantifies the contribution of each input feature to the model's output.

[17]RankIC measures the model's predictive power for delta-neutral returns and is less sensitive to extreme values compared to IC.

[18]In Panel C of Figure 2, the average SHAP value of implied volatility is normalized to 1. By contrast, the most important technical indicator, *rollingstd*, has an average importance of only 0.19, less than one-fifth that of implied volatility.

[19]Similar results can also be found in characteristics from other groups, such as *lso* in Figure 4c and *pcpv30* in Figure 4f.

[20]Demand pressure is scaled by historical mean to ensure comparability across different underlying assets.

[21]Supporting Information S1: Appendix H presents additional analyses of interaction effect. For example, we explore the relationship between expected returns and other representative option factors at different levels of delta.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.